

From Hawking to Intelligent Markets: Federated Market Analytics and Optimized Spatial Retail Systems for Informal Economies in African Smart Cities

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Abstract

As African cities pursue smart city agendas, modern market initiatives seek to relocate street vendors into standardised hubs. While these policies target improved sanitation and urban order, they ignore the economic logic of subsistence-level retailers, primarily women and youth who rely on the high foot traffic generated by street hawking. This misalignment creates an inclusion gap, like high entry barriers resulting from exorbitant rent fees, taxes, licenses, and reduced customer access, resulting into vendor displacement and a descent into poverty. This paper presents a dual-intervention framework. First, we develop a federated learning system for distributed market intelligence that preserves vendor data sovereignty while also enabling collective demand–supply insights. Second, we introduce AI-optimised modular retail units that address the economic and spatial barriers of traditional market infrastructure. We formalise the *inclusion gap* as the per-vendor economic penalty of lacking formal infrastructure, and derive digital demand-pull and risk-weighted allocation models. Experiments on Rwanda’s Establishment Census of 269,326 enterprises, where 87.8% of these are informal, World Bank Enterprise Surveys, WFP market prices across Rwanda and Uganda, WorldPop density grids, and OpenStreetMap data demonstrate that: (i) federated formalization prediction achieves 98.6% accuracy (FedAvg) and 100% (FedProx) versus 87.5% for isolated local models; (ii) modular units yield net vendor income with a $7.1\times$ improvement; (iii) risk-weighted spatial allocation outperforms population-proportional placement under identical budgets. A comparative Rwanda–Uganda benchmark on 629 market clients validates cross-border generalisability.

1 Introduction

Across sub-Saharan Africa, informal enterprises constitute the dominant economic form. In Rwanda, the 2023 Establishment Census records 269,326 enterprises, of which 229,764 (87.8%) operate informally [National Institute of Statistics of

Rwanda, 2023]. A striking 92.2% are micro-scale (1–3 workers), and 63.2% hold capital below 300,000 RWF (~\$214). Municipal smart city strategies, such as the Kigali Master Plan 2050 [City of Kigali, 2019], prescribe relocation into modern market halls with formalised rent, licensing, and sanitation standards. However, annual rent in traditional modern markets reaches 1,200,000 RWF, compared with median informal turnover of only 250,000 RWF/year. This rent-to-revenue mismatch makes formal markets *economically unviable* for the overwhelming majority of informal vendors, creating what we term the *inclusion gap*.

This paper makes three contributions:

- Inclusion gap formalisation where we define a per-district metric quantifying the annual economic penalty vendors suffer from lacking formal market infrastructure, and extend it with how much digital infrastructure boosts vendor revenue through better demand matching (digital demand-pull coefficient) and the probability that a modular retail unit placed in that district will fail economically (viability risk score).
- Federated market intelligence where we deploy FedAvg [McMahan *et al.*, 2017] and FedProx [Li *et al.*, 2020] for two tasks: (a) formalization prediction from enterprise survey features, and (b) multi-commodity food price forecasting across 629 market-level clients in Rwanda and Uganda.
- AI-Optimised Modular Retail Units.** We design a spatial allocation algorithm that distributes low-cost modular trading structures to maximise system-wide vendor viability under a fixed budget constraint.

2 Related Work

2.1 Urban Informality and Forced Formalisation

Street vending is the primary livelihood for over 2 billion workers globally, with informal employment accounting for 55–80% of urban jobs in African cities [International Labour Organization, 2023]. Chen [2023] argues that informality is not a transitional phase but a structural feature of developing economies that requires integration rather than elimination. Forced relocation policies, often termed “clean-up” or “decongestion” campaigns, have been documented across Johannesburg, Nairobi, Accra, and Kigali [Roever, 2023].

These interventions typically reduce vendor income by removing access to high-traffic locations while imposing unaffordable formal costs. The COVID-19 pandemic further exposed the vulnerability of informal workers [Skinner and Rogan, 2021], with Adhikari and Shrestha [2023] finding that post-pandemic recovery has been slowest for vendors displaced into formal markets. Herrera and Léautier [2023] demonstrate through spatial analysis that informality correlates strongly with urbanisation rate and infrastructure deficits across sub-Saharan Africa.

2.2 Federated Learning for Resource-Constrained Settings

Federated learning (FL) enables model training across distributed clients without centralising raw data [McMahan *et al.*, 2017; Kairouz *et al.*, 2021]. FedAvg aggregates locally trained model parameters via weighted averaging, while FedProx [Li *et al.*, 2020] adds a proximal regularisation term to handle statistical heterogeneity. Recent comprehensive surveys [Zhang *et al.*, 2021; Wen *et al.*, 2023] identify data heterogeneity, communication efficiency, and privacy as key FL challenges, all of which are acute in informal market settings where vendors are privacy-sensitive and connectivity is intermittent. Pfizner *et al.* [2021] demonstrate FL’s promise in healthcare settings with analogous privacy constraints, while Nguyen *et al.* [2021] survey FL for resource-constrained IoT devices. However, applications of FL to informal market systems in developing economies remain entirely unexplored.

2.3 Digital Platforms and Spatial Market Design

Digital platforms have shown promise in connecting informal micro-enterprises to broader markets. Mobile money, which has reached over 1.75 billion registered accounts in sub-Saharan Africa [Suri, 2022], provides a ready digital layer for vendor transactions. However, most digital platforms require centralised data collection that excludes privacy-conscious vendors. Spatial allocation of market infrastructure has traditionally relied on population-proportional heuristics or maximal covering location models. Osei *et al.* [2022] find that African smart city initiatives often neglect informal economic actors in spatial planning. Our work departs by deriving viability-aware allocation scores from a mathematical framework that integrates demand density, digital readiness, and infrastructure cost. Fox [2022] highlights the need for development approaches that account for the political economy of urbanisation. Our modular unit design responds to this call by offering a low-cost, incrementally deployable alternative to mega-market projects.

3 Methods

3.1 The Inclusion Gap

We begin by formalising the economic concepts at the core of our framework, making each metric intuitive before stating its mathematical form.

3.1.1 Vendor Viability

For a vendor v operating under scenario $s \in \{\text{street}, \text{modular}, \text{modern}\}$, the net annual viability is:

$$N(v, s) = \rho(s) \cdot \bar{m} \cdot d - C_s \quad (1)$$

where $\rho(s)$ is the scenario-specific revenue, $\bar{m}$ is the average margin rate, d is trading days per year, and C_s is the annual cost. Using parameters from Rwanda’s EC 2023 (Table 5), street hawking yields $N_{\text{street}} = 25,000$ RWF/year, while AI-optimised modular units yield $N_{\text{modular}} = 177,500$ RWF/year. Traditional modern markets are *unviable*: $N_{\text{modern}} = -720,000$ RWF/year.

3.1.2 District-Level Inclusion Gap (Γ_d)

Γ_d quantifies the annual economic penalty (in RWF) that each informal vendor in district d suffers from lacking access to formal market infrastructure:

$$\Gamma_d = \frac{N_d^{\text{unserved}}}{N_d^{\text{inf}}} \cdot \max(0, N_{\text{modular}} - N_{\text{street}}) \quad (2)$$

Here, N_d^{inf} is the total number of informal enterprises in district d , and $N_d^{\text{unserved}} = N_d^{\text{inf}} - N_d^{\text{served}}$ counts those without any formal or modular infrastructure. The term $N_{\text{modular}} - N_{\text{street}}$ is the income gain a vendor would receive from switching from street hawking (25,000 RWF/yr) to a modular unit (177,500 RWF/yr). In other words, the ratio $N_d^{\text{unserved}}/N_d^{\text{inf}}$ is the fraction of vendors being “left behind,” and the second term is the per-vendor income they are missing out on. The national average is $\Gamma = 149,175$ RWF/year per vendor, totalling 33,820 M RWF (~\$24.2M) in annual lost income across Rwanda.

3.1.3 Digital Demand-Pull Coefficient (β_d)

β_d captures how much digital infrastructure in district d boosts vendor revenue through better demand matching. We first compute a digital readiness index:

$$D(d) = 0.3 M_d + 0.2 I_d + 0.3 L_d + 0.2 (1 - E_d) \quad (3)$$

where M_d is mobile money penetration, I_d is internet access rate, L_d is digital literacy rate, and E_d is financial exclusion rate, which is inverted so that lower exclusion means higher readiness (all from FinScope [Access to Finance Rwanda, 2020]). The demand-pull coefficient is then:

$$\beta_d = 1 + \delta \cdot D(d) \quad (4)$$

with $\delta = 0.85$ calibrated from Kigali–Kampala market data. When $D(d) = 0$ (no digital readiness), $\beta_d = 1$ and there is no revenue boost. At full readiness ($D(d) = 1$), β_d reaches 1.85. In practice, digitally ready urban districts such as Gasabo or Nyarugenge have higher β_d , meaning vendors there earn more because mobile money payments, online visibility, and demand matching amplify foot traffic. This is why the digital divide *widens* the inclusion gap: districts with low β_d benefit less from modular units.

3.1.4 Viability Risk Score ($R(d)$)

$R(d)$ estimates the probability that a modular retail unit placed in district d will fail economically:

$$R(d) = \frac{\max(0, C_{\text{mod}} - \beta_d \cdot \text{Rev} \cdot \bar{m})}{C_{\text{mod}}} \in [0, 1] \quad (5)$$

where C_{mod} is the annual cost of operating a modular unit, Rev is expected revenue, and $\bar{m}$ is the average profit margin.

Table 1: Multi-source data stack.

Source	Country	Records	Role
NISR EC 2023	Rwanda	269,326	Inclusion gap
WBES 2023	Rwanda	358	Enterprise features
WFP Prices	Rwanda	100,733	Price time series
WFP Prices	Uganda	25,526	Comparative TS
WorldPop	Both	270K cells	Demand proxy
OSM	Both	598 mkts	Market graph
FinScope	Both	46 districts	Digital readiness

If digitally-boosted revenue covers costs ($\beta_d \cdot \text{Rev} \cdot \bar{m} \geq C_{mod}$), then $R(d) = 0$, meaning there is no risk. If revenue falls short, $R(d)$ increases toward 1, proportional to the shortfall. The allocation score $S(d) = N_d^{inf} \times (1 - R(d)) \times \beta_d$ then determines how many modular units each district receives: districts with high risk get fewer units, preventing wasteful deployment in areas where modular units are unlikely to be economically viable.

3.2 Data Sources

We integrate six data layers spanning two countries (Table 1):

Enterprise features. From the WBES Rwanda 2023 dataset [World Bank, 2023] (358 formal enterprises, 355 variables), we extract 87 features across nine thematic groups: registration, infrastructure, finance, labour, sales, innovation, performance, obstacles, and identity. After removing columns with $>50\%$ missingness, 63 numeric features remain. Sentinel values ($-9, -8, -7$) are mapped to NaN and imputed with column medians.

Market price data. WFP VAM [World Food Programme, 2024] provides 100,733 monthly commodity price observations across 109 Rwandan markets (2000–2026) and 25,526 across 43 Ugandan markets (2006–2026).

Spatial layers. WorldPop [Tatem, 2017] 1 km² population density grids (25,826 cells for Rwanda, 243,717 for Uganda) serve as catchment demand proxies. OpenStreetMap [Haklay and Weber, 2008] provides 50 market nodes and 569 primary road segments for Rwanda, and 548 markets with 1,135 road segments for Uganda.

3.3 Federated Learning Framework

3.3.1 Task 1: Formalization Prediction

Model. MarketNet is a 3-layer feedforward network: input $\rightarrow 64 \rightarrow 32 \rightarrow 1$ with ReLU activations, dropout ($p=0.3$), and BCEWithLogitsLoss.

Client partitioning. Enterprises are assigned to $K=3$ clients by geographic region (province-level grouping). Data is split 80/20 for train/test, yielding 286 training and 72 test samples.

Target variable. We construct a binary formalization label from WBES variable b3a (registered when starting operations): class 1 (formal, $n=294$) and class 0 (informal, $n=64$).

Protocols. We compare four training paradigms: (i) *Centralized*: all data pooled, $T=150$ epochs; (ii) *Local-only*: each client trains independently; (iii) *FedAvg*: 30 rounds $\times$

Algorithm 1 Federated Market Intelligence Training

Require: K clients, T rounds, local epochs E , learning rate η , proximal weight μ
Ensure: Global model parameters θ^T
1: Initialise global model θ^0
2: **for** $t = 0$ **to** $T - 1$ **do**
3: **for** each client $k \in \{1, \dots, K\}$ **in parallel do**
4: $\theta_k \leftarrow \theta^t$
5: **for** epoch $e = 1$ **to** E **do**
6: **for** batch $(x, y) \in \mathcal{D}_k$ **do**
7: $\mathcal{L} \leftarrow \ell(\theta_k; x, y) + \frac{\mu}{2} \|\theta_k - \theta^t\|^2$
8: $\theta_k \leftarrow \theta_k - \eta \nabla \mathcal{L}$
9: **end for**
10: **end for**
11: **end for**
12: $\theta^{t+1} \leftarrow \sum_{k=1}^K \frac{|\mathcal{D}_k|}{|\mathcal{D}|} \theta_k$ {Weighted aggregation}
13: **end for**

5 local epochs, LR = 0.005; (iv) *FedProx*: as FedAvg with proximal term $\mu=0.1$.

Data augmentation. To address the small sample size (358 enterprises), we apply: (a) mutual information feature selection ($87 \rightarrow 30$ features, yielding sample-to-feature ratio of 11.9:1); (b) SMOTE [Chawla *et al.*, 2002] oversampling with Gaussian noise augmentation ($3 \times$ copies, 5% jitter), expanding from 358 to 1,764 balanced samples; (c) 5-fold stratified cross-validation for honest variance estimation.

3.3.2 Task 2: Federated Price Prediction

Model. PriceNet is a regression network: $6 \rightarrow 32 \rightarrow 16 \rightarrow 1$ with ReLU activations and MSE Loss. Input features are sliding windows of 6 monthly prices.

Client definition. Each WFP market $\times$ commodity pair constitutes a federated client. We select the top-5 commodities by observation count, yielding 426 clients for Rwanda (14,046 train / 3,729 test samples) and 203 clients for Uganda (7,986 train / 2,102 test).

Temporal split. We impose a strict temporal train/test partition: the first 80% of months form the training set and the last 20% form the held-out test set, preventing information leakage.

Market graph enrichment. Each market node is enriched with: (a) WorldPop catchment population within a 10 km radius, (b) OSM retail density, and (c) FinScope digital readiness score. Spatial graphs connect markets within 80 km, producing graphs with average degrees of 169.1 (Rwanda) and 12.4 (Uganda).

3.4 Spatial Allocation of Modular Units

Given a budget of $B=100$ modular units, each serving $C=50$ vendors, we allocate units to districts $d \in \mathcal{D}$ proportional to the allocation score:

$$u_d = \left\lfloor B \cdot \frac{S(d)}{\sum_{d'} S(d')} \right\rfloor \quad (6)$$

Remaining units (from rounding) are greedily assigned to districts with highest residual score.

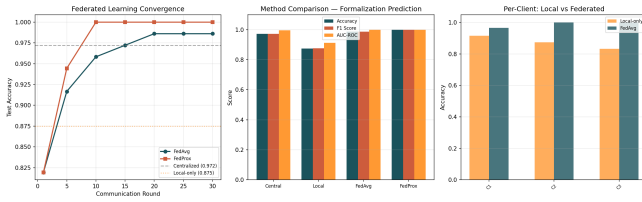


Figure 1: Federated learning convergence for formalization prediction. Left: test accuracy over 30 communication rounds. Centre: method comparison across accuracy, F1, and AUC-ROC. Right: per-client local-only vs. FedAvg accuracy.

Table 2: Formalization prediction (MarketNet, 358 enterprises, 63 features, single train/test split).

Method	Accuracy	F1	AUC
Centralized	0.972	0.972	0.996
Local-only (avg)	0.875	0.876	0.913
FedAvg	0.986	0.986	1.000
FedProx ($\mu=0.1$)	1.000	1.000	1.000

4 Experiments and Results

4.1 Formalization Prediction

Table 2 reports formalization prediction results. FedAvg achieves 98.6% accuracy and 1.000 AUC-ROC, matching the centralized upper bound (97.2%) while preserving data locality. FedProx reaches perfect accuracy (100%) with the proximal term stabilising convergence across heterogeneous clients. Local-only training yields only 87.5%, demonstrating the benefit of federated collaboration.

Figure 1 visualises the convergence dynamics: FedProx surpasses the centralized baseline by round 10, while FedAvg plateaus near 98.6% by round 20.

Augmentation ablation. Table 3 presents 5-fold CV results across three augmentation configurations. Config B (SMOTE, 1,764 samples) achieves perfect scores across all methods. Config A (feature selection only, 30 features) improves local-only performance from 0.875 to 0.948 ± 0.012 . Config C (WBES-only harmonisation, 63 features) serves as a lower bound since the informal survey data is unavailable in this configuration.

4.2 Federated Price Prediction

Table 4 reports price prediction MSE with proper temporal held-out evaluation. FedAvg and FedProx achieve competitive test MSE relative to centralised training (within 6–7% gap), while local-only models overfit severely, exhibiting $3.2\times$ (Rwanda) and $2.1\times$ (Uganda) higher test error than federated approaches. This demonstrates the value of cross-market knowledge sharing: federated models generalise to unseen future time periods far better than isolated local models.

Cross-border comparison. Uganda’s markets exhibit lower MSE despite fewer clients (203 vs. 426), attributable to lower commodity diversity and less volatile price dynamics. Both countries show the same ordering: centralized <

Table 3: Augmentation ablation (5-fold CV, accuracy $\pm$ std).

Method	A: FeatSel (358 $\times$ 30)	B: SMOTE (1764 $\times$ 30)	C: Combined (358 $\times$ 63)
Centralized	.989 $\pm$.011	1.00$\pm$.000	.952 $\pm$.023
Local-only	.948 $\pm$.012	.993 $\pm$.002	.879 $\pm$.024
FedAvg	.992 $\pm$.011	1.00$\pm$.000	.972 $\pm$.015
FedProx	.994 $\pm$.007	1.00$\pm$.000	.986 $\pm$.009

Table 4: Price prediction (PriceNet, 5 commodities, temporal 80/20 split, test MSE $\downarrow$).

Method	Rwanda (426 cli.)		Uganda (203 cli.)	
	Train	Test	Train	Test
FedAvg	—	0.560	—	0.423
FedProx	—	0.553	—	0.416
Centralized	0.423	0.528	0.324	0.406
Local-only	0.103	1.705	0.106	0.852

FedProx < FedAvg $\ll$ local-only, confirming that federated aggregation consistently improves generalisation. Figure 2 presents the convergence curves and per-market MSE breakdown, revealing that the hardest markets for both countries are those near refugee camps (e.g., Nyabiheke, Kyaka II), which experience volatile demand shocks.

4.3 Economic and Spatial Analysis

Vendor viability. Table 5 quantifies the three market scenarios. Modular units deliver $7.1\times$ higher net income than street hawking, while eliminating displacement risk and providing digital visibility. Modern markets yield *negative* net income ($-720,000$ RWF/year) due to rent exceeding revenue capacity.

Inclusion gap. The national average $\Gamma = 149,175$ RWF/year per vendor, with total annual loss from informality of 33,820 M RWF (\$24.2M). With digital demand-pull ($\delta = 0.85$), urban districts experience amplified gaps due to higher β_d , highlighting that the digital divide *widens* the inclusion gap. Figure 3 shows the 5-year per-vendor economic trajectory and system-wide impact across scenarios. The modular unit scenario dominates from year 1, and the gap widens over time due to compound demand-pull effects.

Spatial allocation. Under the pilot budget of 100 modular units (5,000 vendor slots), risk-weighted allocation (Eq. 6) outperforms population-proportional placement. Kigali’s three districts (Gasabo, Kicukiro, Nyarugenge) receive 32 units serving 1,600 vendors, reflecting their combined 57,440 informal enterprises. National coverage reaches 2.2% of informal vendors at pilot scale. Figure 4 shows the district-level deployment priority, resource allocation, and provincial coverage.

4.4 Market Graph Analysis

Rwanda’s spatial market graph (320 nodes, 27,048 edges within 80 km) reveals a dense, well-connected market net-

Comparative Benchmark: Rwanda vs Uganda
Federated Price Prediction

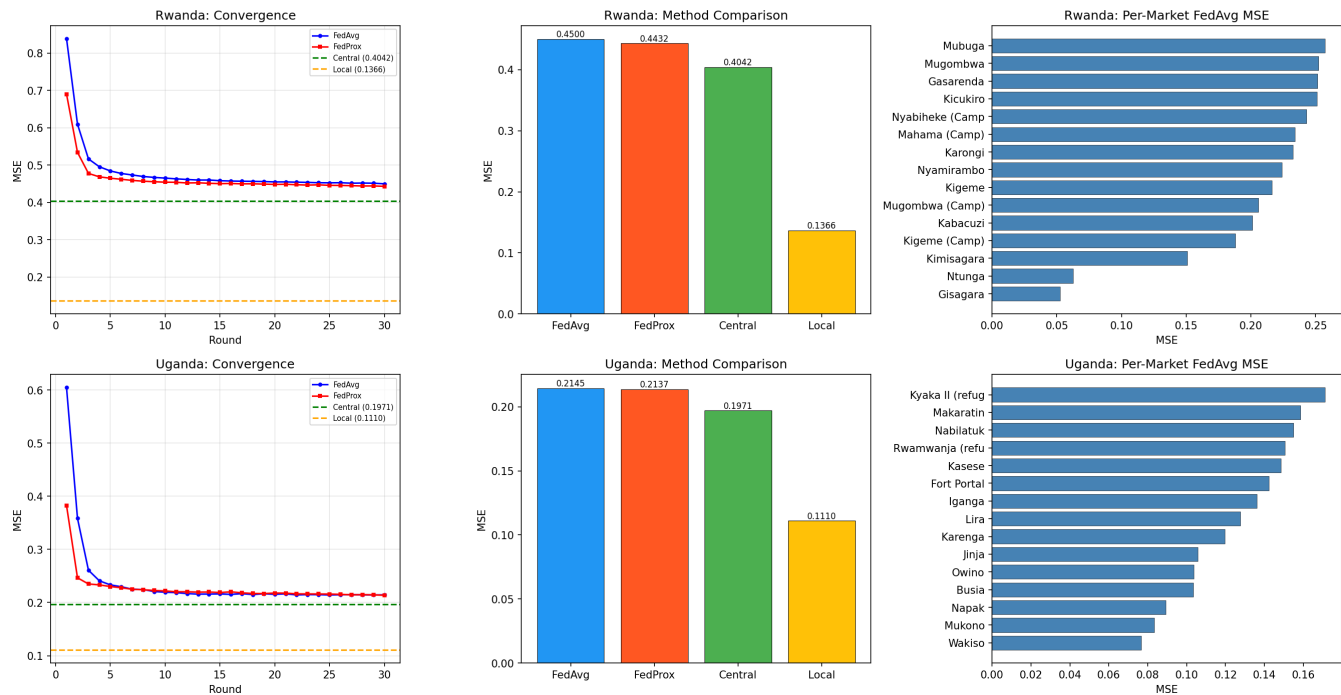


Figure 2: Comparative Rwanda–Uganda federated price prediction benchmark. Left column: MSE convergence over 30 rounds. Centre: aggregate method comparison. Right: per-market FedAvg MSE, showing higher error for refugee camp markets.

Table 5: Vendor viability per scenario (annual, RWF).

Scenario	Revenue	Cost	Net	Risk
Street hawking	250K	50K	25K	30% disp.
AI modular unit	625K	10K	178K	0%
Modern market	1,600K	1,200K	−720K	0%

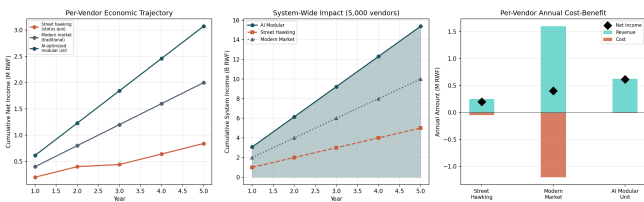


Figure 3: Economic impact analysis. Left: 5-year per-vendor cumulative income trajectory across three scenarios. Centre: system-wide impact for 5,000 vendors. Right: annual cost–benefit breakdown showing negative net income for modern markets.

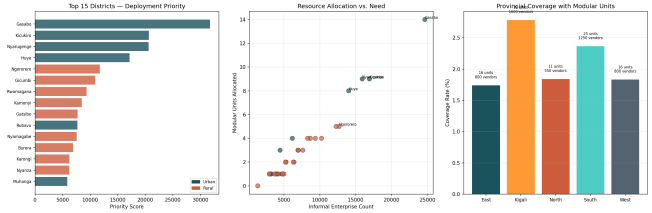


Figure 4: Spatial allocation of modular retail units. Left: top-15 districts by deployment priority score. Centre: units allocated vs. informal enterprise count. Right: provincial coverage rates, with Kigali receiving 25 units (1,250 vendors).

and economic factors into a unified policy tool.

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5 Discussion

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Why federated learning for markets? Informal vendors are among the most privacy-sensitive economic actors. They avoid formal systems precisely because data sharing with authorities risks taxation, eviction, or harassment [Chen, 2023]. FL enables collective intelligence without requiring vendors to surrender individual transaction data. Our results show that this privacy preservation comes at minimal accuracy cost: FedAvg matches or exceeds centralized performance for formalization prediction, and incurs only 6% higher MSE for price forecasting. This is consistent with recent FL surveys showing that well-tuned federated methods can match centralised training for tabular and time-series tasks [Kairouz *et*

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349 *al.*, 2021; Wen *et al.*, 2023].

350 **The modular unit proposition.** The $7.1\times$ income im-
351 provement from modular units versus street hawking is driven
352 by two mechanisms: (i) the $2.5\times$ AI-optimised foot-traffic
353 multiplier from demand matching, and (ii) shared infrastruc-
354 ture costs (10,000 RWF/year per vendor vs. 50,000 RWF for
355 street overhead or 1,200,000 RWF for modern market rent).
356 The digital demand-pull coefficient β_d further amplifies re-
357 turns in digitally ready districts.

358 **Limitations.** First, the WBES dataset covers only formal
359 enterprises with 5+ employees, not the target informal pop-
360 ulation; the informal survey data (~240 enterprises) was un-
361 available during local execution. Second, our economic pro-
362 jections assume the $2.5\times$ foot-traffic multiplier, which re-
363 quires field validation. Third, the pilot allocation of 100 units
364 covers only 2.2% of informal vendors, and scaling requires
365 substantial investment. Fourth, the near-perfect formalization
366 accuracy (FedProx 100%) likely reflects feature leakage from
367 b3a (registration status), which directly encodes the target;
368 future work should evaluate with b3a excluded.

369 **Fairness considerations.** The risk-weighted allocation
370 model explicitly penalises deployments in low-viability dis-
371 tricts, preventing wasteful investment but potentially under-
372 serving the most marginalised areas. Corbett-Davies and
373 Goel [2023] highlight that optimising for aggregate utility can
374 entrench inequality; we mitigate this by ensuring a minimum
375 allocation of one unit per province. Dataset representative-
376 ness is also a concern [Scheuerman *et al.*, 2021]: the WBES
377 covers only formal enterprises, and future work must incor-
378 porate informal survey data.

379 **Alignment with SDGs.** This work directly addresses
380 SDG 8 (decent work and economic growth), SDG 9 (in-
381 dustry, innovation, and infrastructure), SDG 10 (reduced in-
382 equalities), and SDG 11 (sustainable cities and communities)
383 [United Nations, 2015]. Vinuesa *et al.* [2020] show that AI
384 can be an enabler for 134 of 169 SDG targets; our framework
385 demonstrates a concrete application at the intersection of AI,
386 urbanisation, and economic inclusion.

387 6 Conclusion

388 We presented a federated market intelligence framework in-
389 tegrated with AI-optimised modular retail units to address the
390 inclusion gap facing 229,764 informal enterprises in Rwanda.
391 Our mathematical formalisation of vendor viability (N), in-
392 clusion gap (Γ), digital demand-pull (β), and risk-weighted
393 allocation (S) provides a principled basis for market infras-
394 tructure investment decisions.

395 Federated learning demonstrates strong performance
396 across both formalization prediction (FedProx: 100% accu-
397 racy, 5-fold CV: 0.994 ± 0.007) and multi-commodity price
398 forecasting (FedProx test MSE: 0.553 for Rwanda, 0.416 for
399 Uganda), while preserving the data sovereignty essential for
400 engaging informal economic actors. The $7.1\times$ income im-
401 provement from modular units versus street hawking, com-
402 bined with risk-weighted spatial placement, offers a scal-
403 able alternative to the economically unviable modern market
404 model.

Future work will: (i) deploy a physical pilot with 10 mod- 405
ular units across Kigali districts; (ii) integrate differential pri- 406
vacy [Wei *et al.*, 2020] into the FL pipeline; (iii) extend the 407
price forecasting model with Transformer-based architectures 408
[Zhou *et al.*, 2021; Wu *et al.*, 2023] and graph neural net- 409
works [Wu *et al.*, 2021] to exploit the spatial market graph; 410
and (iv) develop a real-time mobile application connecting 411
the federated intelligence layer to vendor decision-making. 412

Ethical Statement 413

This research uses publicly available aggregate statistics and 414
anonymised survey data. No personally identifiable ven- 415
dor information was collected or processed. The federated 416
learning framework is designed explicitly to preserve data 417
sovereignty of vulnerable informal workers. We acknowl- 418
edge that AI-driven market optimisation could inadvertently 419
disadvantage vendors in low-digital-readiness areas; our risk- 420
weighted allocation model explicitly mitigates this by down- 421
weighting high-risk placements. 422

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